

Postdoctoral Researcher at University of Toronto and Vector Institute

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Current Position

University of Toronto & The Vector Institute - *Postdoctoral Researcher*

2024-Present

Advised by [Aleksandar Nikolov](#) and [Nicolas Papernot](#)

Education

The Ohio State University - *PhD in Computer Science*

2019 - 2024

Advised by [Raef Bassily](#)

Case Western Reserve University - *BS in Computer Science*

2015 - 2019

Publications (Author orderings a mix of alphabetical and contribution order. A "*" indicates I was a primary author)

1. ***Stronger Memory-Query Tradeoffs for Convex Optimization: The Limitations of Subquadratic Memories**
Michael Menart, Aleksandar Nikolov, Ohad Shamir, Foundations of Computer Science (FOCS). Nov. 2026.
2. ***On the Gradient Complexity of Private Optimization with Private Oracles**
Michael Menart, Aleksandar Nikolov. Conference on Learning Theory (COLT). June 2026.
3. **Private Rate-Constrained Optimization with Applications to Fair Learning**
Mohammad Yaghini, Tudor Cebere, **Michael Menart**, Aurélien Bellet, Nicolas Papernot. ICLR, 2026.
4. ***Private Algorithms for Stochastic Saddle Points and Variational Inequalities: Beyond Euclidean Geometry**
Raef Bassily, Cristóbal Guzman, Michael Menart. Neural Information Processing Systems (NeurIPS). Dec 2024.
5. ***Public-data Assisted Private Stochastic Optimization: Power and Limitations**
Enayat Ullah, Michael Menart, Raef Bassily, Cristóbal Guzman, Raman Arora. Neural Information Processing Systems (NeurIPS). Dec 2024.
6. ***Differentially Private Non-Convex Optimization under the KL Condition with Optimal Rates**
Michael Menart, Enayat Ullah, Raman Arora, Raef Bassily, Cristóbal Guzman. International Conference on Algorithmic Learning Theory (ALT) 2024.
7. ***Differentially Private Algorithms for the Stochastic Saddle Point Problem with Optimal Rates for the Strong Gap**
Raef Bassily, Cristóbal Guzman, **Michael Menart**. Conference on Learning Theory (COLT). July 2023.
8. ***Faster Rates of Convergence to Stationary Points in Differentially Private Optimization**
Raman Arora, Raef Bassily, Tomás Gonzalez, Cristóbal Guzman, **Michael Menart**, Enayat Ullah. International Conference on Machine Learning (ICML). July 2023.
9. ***Differentially Private Generalized Linear Models Revisited**

Raman Arora, Raef Bassily, Cristóbal Guzman, **Michael Menart**, Enayat Ullah. Neural Information Processing Systems (NeurIPS). Nov. 2022.

10. *Differentially Private Stochastic Optimization: New Results in Convex and Non-Convex Settings

Raef Bassily, Cristóbal Guzman, **Michael Menart**. Conference on Neural Information Processing Systems (NeurIPS). Dec. 2021.

Teaching Experience

Tutorials - <i>miscellaneous tutorial lectures given at Vector Institute</i>	2024-2025
Teaching Assistant - <i>for class "Machine Learning"</i>	Spring 2022
Teaching Assistant - <i>for class "Intro to C++"</i>	Spring 2020
Programming Tutor - <i>individual tutoring for Python</i>	Fall 2018

Work Experience

MIM Software Inc. (Research Intern)	2018-2019 (Summers)
Explored/implemented machine learning methods for medical imaging and NLP applications in Python.	
Air Force Research Laboratory (AFRL) (Research Intern)	2013 - 2017 (Summers)
Implemented various projects in computer vision/deep learning in Python/C++.	

Talks/Presentations

- Beyond Algorithms: Machine Unlearning from a Multidisciplinary Perspective, "Technical Approaches to Mitigating Copyright Infringement in Generative Models", Aug 2026
- SIAM OP26 Minisymposium Talk, "Stochastic Saddle Point Problems under Differential Privacy"
- Foundations of AI Security Symposium (invited tutorial), "Techniques and Bottlenecks in DP Optimization", March 2026
- Vector Remarkable. Privacy and Security Workshop. "DP-SGD Is Necessarily Slow", Feb 2026
- University of Toronto. Theory Seminar. "On the Gradient Complexity of Private Optimization from Private Oracles", Dec 2025
- University of Waterloo. Algorithms and Complexity Seminar, "On the Gradient Complexity of Private Optimization from Private Oracles", Nov 2025
- Vector Institute, Tutorial: Introduction to Non-convex Optimization, Nov 2025
- Discussion moderator, AI Governance Mechanisms Workshop, April 2025
- Vector Institute, Tutorial on Mirror Descent, Feb 2025
- University of Toronto Theory Lunch Talks, "Lower Bound Techniques in Differential Privacy", Nov 2024
- Translational Data Analytics Institute Graduate Seminar Series, "The Role of Public Data in Private Machine Learning: Power and Limitations", 2024
- Boston Area Data Privacy Seminar, "The Complexity of Differentially Private Optimization for Machine Learning", Nov. 2023
- Optimization for Machine Learning Workshop, "Differentially Private Generalized Linear Models Revisited", *International Conference on Continuous Optimization (ICCOPT)*, 2022 (served as Session Chair)

Reviewing

I have served on the program committee or provided reviews for various conferences and journals, including ALT, COLT, NeurIPS, ICML, ICLR, SIAM, SODA, JMLR, IEEE TDSC, TMLR.